
KidsChem

Master Product Requirement & Engineering Architecture Specification

System Title: Interactive Science Simplifier & Visual Sandbox Ecosystem

Regional Market Context: Sierra Leone (Primary & Junior Secondary School Tiers)

Core Knowledge Integration: Aki-Ola Series, MBSSE Pupil Handbooks, Evans, Longman Science

Target Audience Matrix: Ages 8–15 Years Old (Segmented Experience Vectors)

Technology Architecture: Next.js (App Router), Tailwind CSS, Framer Motion, Supabase Core, Monime API

Document Version: 1.0.0-PROD_RELEASE

1. Vision & Strategic Intent

KidsChem is a category-defining interactive edtech ecosystem engineered to dismantle complex text barriers, eliminate textbook anxieties, and replace dry rote memorization patterns with spatial, intuitive frameworks for children aged 8–15 in Sierra Leone. The product directly digests heavyweight, rigorous curricula from dominant regional textbooks—specifically *Aki-Ola Integrated Science*, the *MBSSE National Pupil Handbooks*, *Evans*, and *Longman Science*—and distills them through a structured five-stage mastery architecture loop.

The absolute foundation of the product is its **uncompromised simplification leverage strategy**: extracting heavy biological formulas, chemical transformations, and laws of physics, and packaging them into highly descriptive, short narrative chapters. These narratives lead instantly into a high-end, touch-sensitive **Virtual Lab Box Sandbox Canvas** where reactions are visualized in real-time. Assisted by an integrated context-bounded AI guardian (Sci-Buddy) and an adaptive re-testing mastery quiz module, KidsChem empowers students with a world-class scientific foundation.

1.1 Executive Spa Visual Aesthetic Guidelines

To protect students from cognitive clutter and lower subject intimidation, KidsChem completely discards the noisy, over-stimulating, high-frequency layouts typical of childish edtech applications. It introduces a high-end, therapeutic minimalist aesthetic:

- **Color Foundations:** Pure canvas white backdrops layered with clean neutral warm cream layout boxes (**#FAF9F6**), stark slate text (**#111111**), and light gray container boundaries (**#DCDCD4**). Real-time progress highlights are rendered in single, rich, desaturated emerald tones.
- **Micro-Physics:** Every element interaction, inventory extraction, canvas combination, or score mutation utilizes precise spring mechanics via *Framer Motion* to deliver premium tactile feedback.

2. Unified Ecosystem Matrix

The system balances account controls, learning analytics, and curated assignment tasks across three distinct roles managed via centralized relationship keys:

USER PERSONA	CORE BEHAVIORAL DIRECTIVES	PLATFORM ACCESS BOUNDARIES
Student (Primary & JSS Tiers)	Completes diagnostic questionnaires, reads conceptual stories, performs experiments inside the Lab Box canvas, views video guides, chats with Sci-Buddy, and resolves adaptive mastery quizzes.	Read-only access to allocated level tracks; write access to individual quiz attempts and personalized progress telemetry tables.
Parent (Account Owner)	Creates student sub-profiles, processes subscription micro-billings, reviews progress map metrics, and executes invitations to delegate private tutors.	Full write access to administrative panels, subscription status rows, and teacher relationship parameters. Read-only access to student analytical history feeds.
Teacher / Tutor (Invited)	Onboarded via secure email invitation hooks from the parent. Reviews student performance trends and dynamically curates mandatory assignment modules.	Full write access to the assignment pipeline table. Read-only access to performance logs of delegated student profiles.

3. Core Product Functional Specifications

Epic A: Habit Onboarding & Personalization Trackers

REQ-001: Weakness Diagnostic Wizard

Upon initial profile execution, the student interacts with a clean, spring-animated question track (e.g., *"Which scientific field feels like a puzzle right now: Biology, Chemistry, or Physics?"*). The wizard captures baseline anxieties and logs the diagnostic state.

REQ-002: Dynamic Progressive Mapping

The platform updates the visual nodes of the student's learning map based on onboarding answers, highlighting weak nodes with micro-rewards and specialized badges to incentivize consistent habit creation.

Epic B: The Core 5-Step Learning Loop

REQ-003: High-Leverage Conceptual Stories

Every curriculum module opens with an ultra-simplified story explaining the topic in highly clear terms. These stories keep official scientific definitions (e.g., density formulas or atmospheric components from Aki-Ola) intact but wrap them in accessible narrative metaphors.

REQ-004: The Tactile Virtual Lab Box Sandbox

A sophisticated workspace powered by spatial canvas collision algorithms. Students manually interact with tools and chemical reagents to see immediate physical and descriptive molecular transformations come alive.

Lab Box Use Case Scenario: Oxygen & Combustion Dynamics

The Canvas Interface: A student selects a lit candle, an inversion beaker glass tool, and an active leaf asset on a clean canvas backdrop.

The Interaction: Dropping the beaker tool directly onto the candle triggers a Framer Motion particle drain animation. The candle flame gradually exhausts the confined oxygen particles and puffs out into clean gray smoke elements.

The Remediation: Dropping the leaf asset into the sealed beaker triggers oxygen production, altering the timeline calculations and updating structural on-screen readouts demonstrating real-time chemical properties.

REQ-005: Localized Video Station Channels

An inline media panel that delivers high-definition instruction explaining the physical mechanics of the simulated lab experiment. Video streams utilize adaptive-bitrate compression algorithms to ensure buffer-free playback across unstable network connections.

REQ-006: Adaptive Mastery Quiz Loop

Quizzes pull questions matching the active topic. If a child enters an incorrect option, the platform blocks completion, reveals a **Remedial Marking Session** citing the exact textbook source layer (e.g., *Aki-Ola Page 42*), and serves a newly structured variation of the question to verify conceptual understanding.

| Epic C: High-Equity AI Companion ("Sci-Buddy")

REQ-007: Bounded Textbook Grounding

A floating conversational module running inline with active tasks. The underlying model parses vector embeddings of full-text science volumes, optimizing answers based on the student's profile level.

REQ-008: Off-Topic Guardrail Classifiers

If a user types queries outside the educational spectrum, Sci-Buddy triggers immediate boundaries: *"That sounds interesting, but I'm optimized for science adventures! Let's get back to exploring our active lab board."* Sensitive biological inputs are strictly audited against the standard academic syllabus and answered with pure, protective, age-appropriate scientific facts.

Epic D: Monime Payment Integration & Core Splits

REQ-009: Weekly USSD Mobile Money Hook

Integrates directly with **Monime API** infrastructure to execute seamless, low-friction micro-payments priced at **Le 5 (Old Leones) per week**, pushed directly to the parent's registered mobile wallet.

REQ-010: Feature Tiers Structure

- **Free Tier ("All the Juice"):** Full operational access to the 150-word stories, virtual Lab Box sandboxes, standard quiz nodes, and Video Station items.
- **Pro Tier ("The Exclusive Edge"):** Unlocks conversational Sci-Buddy AI sessions, deep textbook cross-reference searches, custom teacher assignments, and premium rank badges.

4. Technical Database Architecture Design

The complete application state maps across a structured, securely relational Supabase PostgreSQL architecture blueprint detailed below:

```
-- Core Enums Configuration
CREATE TYPE user_role AS ENUM ('parent', 'teacher', 'student');
CREATE TYPE science_tier AS ENUM ('PRIMARY', 'JSS');
CREATE TYPE science_subject AS ENUM ('BIOLOGY', 'CHEMISTRY', 'PHYSICS');
CREATE TYPE payment_status AS ENUM ('PENDING', 'SUCCESSFUL', 'FAILED');

-- Profiles Core Layout Table
CREATE TABLE public.profiles (
    id UUID REFERENCES auth.users ON DELETE CASCADE PRIMARY KEY,
    full_name TEXT NOT NULL,
    role user_role NOT NULL DEFAULT 'parent',
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- Student Telemetry Meta Table
CREATE TABLE public.students (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    parent_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
    full_name TEXT NOT NULL,
    tier science_tier NOT NULL,
    targeted_weakness science_subject,
    continuous_streak_days INT DEFAULT 0 NOT NULL,
    avatar_url TEXT,
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- Premium Monime Subscription Control Ledger
CREATE TABLE public.subscriptions (
```

```

    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    parent_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
    is_pro_active BOOLEAN DEFAULT FALSE NOT NULL,
    tier_price_leones NUMERIC(10, 2) DEFAULT 5.00 NOT NULL,
    period_start TIMESTAMPTZ DEFAULT NOW() NOT NULL,
    period_end TIMESTAMPTZ NOT NULL,
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- Monime API Live Transaction Registry
CREATE TABLE public.monime_transactions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    parent_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
    subscription_id UUID REFERENCES public.subscriptions(id) ON DELETE SET NULL,
    amount NUMERIC(10, 2) NOT NULL,
    phone_number TEXT NOT NULL,
    monime_reference TEXT UNIQUE NOT NULL,
    status payment_status NOT NULL DEFAULT 'PENDING',
    webhook_raw_payload JSONB,
    created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL
);

-- Master Science Syllabus Reference Table
CREATE TABLE public.topics (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    subject science_subject NOT NULL,
    tier science_tier NOT NULL,
    title TEXT NOT NULL,
    slug TEXT UNIQUE NOT NULL,
    story_content TEXT NOT NULL,
    video_url TEXT,
    lab_config JSONB NOT NULL, -- Declarative canvas matrix configuration
    sort_order INT NOT NULL
);

-- Quiz Definition Storage
CREATE TABLE public.quizzes (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    topic_id UUID REFERENCES public.topics(id) ON DELETE CASCADE UNIQUE NOT NULL,
    passing_score INT DEFAULT 80 NOT NULL
);

-- Itemized Question Core Repository
CREATE TABLE public.quiz_questions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    quiz_id UUID REFERENCES public.quizzes(id) ON DELETE CASCADE NOT NULL,
    question_text TEXT NOT NULL,
    options JSONB NOT NULL,
    correct_option TEXT NOT NULL,
    explanations_from_textbook TEXT NOT NULL
);

-- Adaptive Remedial Loop Live Tracker Table
CREATE TABLE public.quiz_attempts (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id UUID REFERENCES public.students(id) ON DELETE CASCADE NOT NULL,
    quiz_id UUID REFERENCES public.quizzes(id) ON DELETE CASCADE NOT NULL,

```

```
question_id UUID REFERENCES public.quiz_questions(id) ON DELETE CASCADE NOT NULL,  
selected_option TEXT NOT NULL,  
is_correct BOOLEAN NOT NULL,  
attempt_iteration INT DEFAULT 1 NOT NULL, -- Tracks current re-test loop index  
created_at TIMESTAMPTZ DEFAULT NOW() NOT NULL  
);
```

5. Technical Security & Processing Safeguards

To preserve state integrity under unpredictable client dropouts, local caching states manage user progress inside browser-bound storage parameters, dispatching small JSON synchronization updates back to Supabase endpoints once a network connection settles. API interactions mapping directly into the Monime infrastructure route through signed cryptographic webhooks to guarantee zero transaction bypass states.